**STUDYNEST | CLASS 11 PHYSICS**

# QUESTION PAPER 1

## Chapter 3: Motion in a Plane

**Maximum Marks: 40 | Suggested time: 60 minutes**

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

### Section A - 1-mark questions - 8 x 1 = 8 marks

Is displacement scalar or vector? [1]
x-component at angle theta? [1]
Horizontal acceleration of an ideal projectile? [1]
Direction of centripetal acceleration? [1]
Magnitude of perpendicular components 3 and 4? [1]
Relation between v and omega? [1]
Ideal maximum-range angle at equal heights? [1]
Vertical velocity at projectile peak? [1]

### Section B - 2-mark questions - 6 x 2 = 12 marks

Resolve 10 N at 60 deg . [2]
Displacement after a full circular lap? [2]
v=12 m/s, r=6 m: a_c? [2]
Horizontal launch from 20 m, g=10: flight time? [2]
Perpendicular vectors 6 and 8: resultant? [2]
120 rpm in rad/s? [2]

### Section C - 3-mark questions - 4 x 3 = 12 marks

Projectile at 20 m/s, 30 deg , g=10: T,H? [3]
Forces 8 N east and 6 N north: resultant? [3]
Cyclist at 10 m/s, radius 25 m: a_c? [3]
25 m/s at 37 deg , sin=.6, cos=.8, g=10: time/range? [3]

### Section D - 4-mark questions - 2 x 4 = 8 marks

Projectile 20 m/s at 30 deg : time, height, range? [4]
5 kg at 4 m/s in radius 2 m: inward force and work? [4]

End of paper - check working, units, signs and significant figures.